

Measurement (8th std)

1. Physics is mainly concerned with understanding:

- A) Living organisms
- B) Physical environment
- C) Chemical reactions
- D) Space travel

Answer: B

2. Physics laws are based on:

- A) Imagination
- B) Guesswork
- C) Observation and experiment
- D) Stories

Answer: C

3. Physics is called the science of:

- A) Motion
- B) Energy
- C) Measurements
- D) Matter

Answer: C

4. Measurement is a comparison of:

- A) Two unknown quantities
- B) Unknown with known quantity
- C) Two known quantities
- D) Instruments

Answer: B

5. The value obtained after measurement is called:

- A) Unit
- B) Magnitude
- C) Scale
- D) Error

Answer: B

6. Magnitude is expressed in:

- A) Only numbers
- B) Only units
- C) Numbers and units
- D) Symbols

Answer: C

7. Length is measured using:

- A) Clock
- B) Balance
- C) Ruler
- D) Thermometer

Answer: C

8. Time is measured using:

- A) Ruler
- B) Clock
- C) Thermometer
- D) Balance

Answer: B

9. Mass is measured using:

- A) Thermometer

- B) Balance
- C) Stopwatch
- D) Meter scale

Answer: B

10. Temperature is measured using:

- A) Balance
- B) Clock
- C) Thermometer
- D) Ammeter

Answer: C

11. A unit is a:

- A) Device
- B) Standard for measurement
- C) Quantity
- D) Error

Answer: B

12. For perfect measurement, we need:

- A) Instrument only
- B) Unit only
- C) Instrument, standard quantity, unit
- D) None

Answer: C

13. SI system stands for:

- A) Standard International
- B) Scientific Index
- C) International System of Units

D) System International

Answer: C

14. SI units are used to:

A) Confuse measurements

B) Standardize measurements

C) Increase errors

D) Reduce numbers

Answer: B

15. SI units are divided into:

A) Two types

B) Three types

C) Four types

D) Five types

Answer: A

16. Fundamental quantities are:

A) Dependent

B) Independent

C) Derived

D) Combined

Answer: B

17. Example of fundamental quantity:

A) Speed

B) Force

C) Length

D) Energy

Answer: C

18. Derived quantities are:

- A) Independent
- B) Basic
- C) Combination of fundamental quantities
- D) Simple

Answer: C

19. Example of derived quantity:

- A) Mass
- B) Time
- C) Speed
- D) Temperature

Answer: C

20. Temperature indicates:

- A) Weight
- B) Hotness and coldness
- C) Speed
- D) Length

Answer: B

21. SI unit of temperature is:

- A) Celsius
- B) Kelvin
- C) Fahrenheit
- D) Joule

Answer: B

22. Electric current is the flow of:

- A) Heat

B) Electricity

C) Light

D) Mass

Answer: B

23. SI unit of electric current is:

A) Volt

B) Ampere

C) Ohm

D) Watt

Answer: B

24. Electric charge is measured in:

A) Ampere

B) Coulomb

C) Volt

D) Joule

Answer: B

25. One ampere is equal to:

A) One coulomb per second

B) One volt per second

C) One joule per second

D) One watt per second

Answer: A

26. Electric current is measured using:

A) Voltmeter

B) Ammeter

C) Thermometer

D) Barometer

Answer: B

27.Amount of substance is measured in:

A) Gram

B) Mole

C) Liter

D) Newton

Answer: B

28.Amount of substance depends on:

A) Size

B) Number of particles

C) Volume

D) Temperature

Answer: B

29.Luminous intensity measures:

A) Heat

B) Sound

C) Light power

D) Pressure

Answer: C

30.SI unit of luminous intensity is:

A) Lumen

B) Candela

C) Watt

D) Lux

Answer: B

31. Luminous intensity is measured using:

- A) Ammeter
- B) Thermometer
- C) Photometer
- D) Barometer

Answer: C

32. SI unit of luminous flux is:

- A) Candela
- B) Lumen
- C) Watt
- D) Joule

Answer: B

33. Plane angle is measured in:

- A) Steradian
- B) Radian
- C) Degree
- D) Lumen

Answer: B

34. Solid angle is measured in:

- A) Radian
- B) Degree
- C) Steradian
- D) Watt

Answer: C

35. Radian is defined using:

- A) Area

- B) Volume
- C) Arc length equal to radius
- D) Diameter

Answer: C

36. Solid angle is:

- A) 2D angle
- B) 3D angle
- C) Linear measure
- D) Area

Answer: B

37. A clock is used to measure:

- A) Length
- B) Time
- C) Mass
- D) Temperature

Answer: B

38. Analog clocks show time using:

- A) Numbers only
- B) Moving hands
- C) Light
- D) Sound

Answer: B

39. Digital clocks display time in:

- A) Hands
- B) Symbols
- C) Numerals

D) Circles

Answer: C

40. Quartz clocks use:

A) Water

B) Sand

C) Quartz crystal

D) Metal

Answer: C

41. Quartz clocks are more accurate than:

A) Digital clocks

B) Mechanical clocks

C) Atomic clocks

D) Solar clocks

Answer: B

42. Atomic clocks are used in:

A) Schools

B) GPS systems

C) Homes

D) Shops

Answer: B

43. GMT is measured at:

A) India

B) USA

C) Greenwich

D) China

Answer: C

44. Earth is divided into how many time zones?

A) 12

B) 24

C) 36

D) 48

Answer: B

45. Time difference between two zones is:

A) 30 minutes

B) 1 hour

C) 2 hours

D) 15 minutes

Answer: B

46. IST is based on longitude:

A) 90°

B) 82.5°

C) 75°

D) 60°

Answer: B

47. IST is ahead of GMT by:

A) 3 hours

B) 4 hours

C) 5:30 hours

D) 6 hours

Answer: C

48. Accuracy means:

A) Repeating values

B) Closeness to true value

C) Large value

D) Small value

Answer: B

49. Precision means:

A) Correct value

B) Repeated same results

C) Large error

D) Approximation

Answer: B

50. Measurement always contains:

A) Zero

B) Error

C) Heat

D) Light

Answer: B

51. Approximation means:

A) Exact value

B) Rough value close to exact

C) Wrong value

D) Large value

Answer: B

52. Rounding off makes numbers:

A) Complex

B) Simpler

C) Larger

D) Smaller always

Answer: B

53.If next digit is less than 5, rounding:

A) Increase

B) Decrease

C) Remains same

D) Multiply

Answer: C

54.If next digit is 5 or more, rounding:

A) Decrease

B) Remains same

C) Increase by one

D) Divide

Answer: C

55.73 rounded to nearest ten is:

A) 60

B) 70

C) 80

D) 90

Answer: B

56.76 rounded to nearest ten is:

A) 60

B) 70

C) 80

D) 90

Answer: C

57. Precision depends on:

- A) Instrument
- B) Environment
- C) Repetition
- D) All

Answer: D

58. SI units help to:

- A) Increase confusion
- B) Avoid confusion
- C) Reduce measurements
- D) Increase errors

Answer: B

59. Photometer measures:

- A) Current
- B) Temperature
- C) Light intensity
- D) Pressure

Answer: C

60. Ammeter measures:

- A) Voltage
- B) Current
- C) Resistance
- D) Power

Answer: B